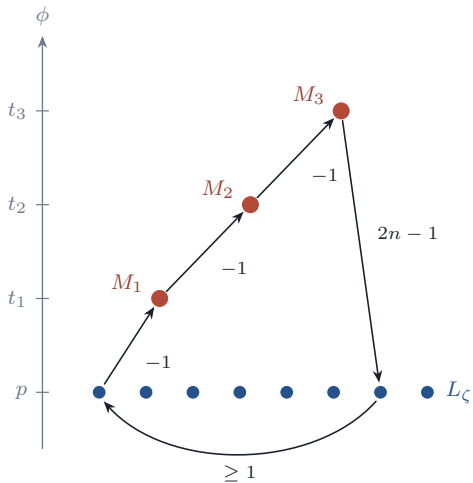
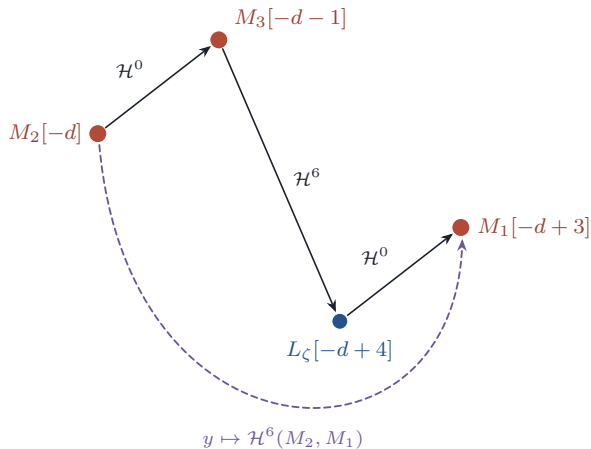


(a) the excess of a cycle,  $n \geq 3$



excess of a step of degree one:  $-1$  up ( $\text{Ext}^0$ ),  
 $2n-1$  down ( $\text{Ext}^{2n}$ ), at least 1 from  $L_\zeta$  to  $L_{\zeta'}$ ,  
 this cycle:  $3 \cdot (-1) + (2n-1) + 1 = 2n-3 > 0$ ;  
 reaching  $H^2(\mathcal{O}_A)$  needs a cycle of excess 0

(b) the one path that acts,  $n = 3$



$y \mapsto \mathcal{H}^6(M_2, M_1)$   
 Ext degrees 0, 6, 0, excesses  $-1, 5, -1$ : the product with  
 $y \in H^2(\mathcal{O}_A)$  lands in degree 6, in a space of  
 dimension  $(t_2 - t_1)^6$ ; the shifts force  
 $\sigma_1 = \sigma_3 = -\sigma_2$